

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	29.08.26

Code of Conduct

I Sania Herbert (192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Rubrics for Calculation ASSESSMENT RUBRIC
AT1 — Uncertainty Visualization Lab
CO5 · BT4: Analyze ·

This rubric is used to assess student responses to the CO5-AT1 Uncertainty Visualization Lab problems, in which students construct and interpret uncertainty visualizations (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, and axis scale choice) from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant uncertainty-visualization technique (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, or axis scale choice) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, including all required calculations.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and calculation techniques to	Applies techniques to produce a correct visualization, with	Limited application; the visualization partially	Fails to apply appropriate techniques; the visualization does

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		produce a complete, accurate uncertainty visualization correctly matched to the data.	only minor errors.	represents the uncertainty but has notable errors.	not correctly represent the uncertainty.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or sequence.	Poorly organized; the response is incoherent or key parts are left unaddressed.
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 21

Data: monthly app downloads (in hundreds) over 6 periods: 41, 49, 59, 56, 61,

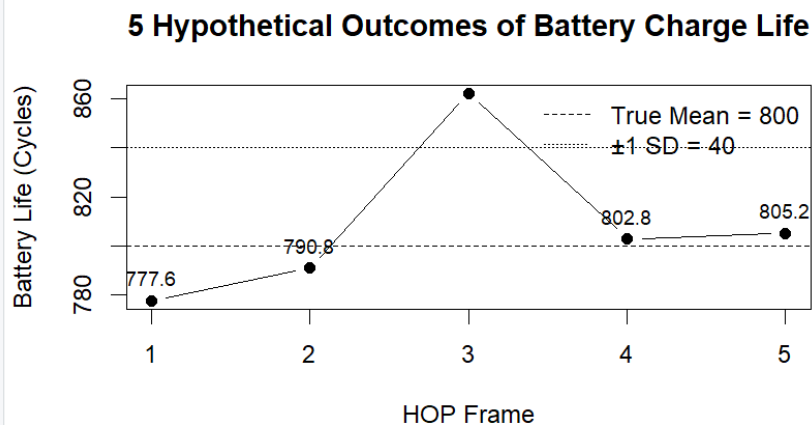
65.

Task: Fit a linear trend line, compute the residuals and their standard deviation, and state the 95% uncertainty band width around the trend.

Python code:

```
1 # Problem 22: Hypothetical Outcome Plot (HOP)
2 # Given population parameters
3 mu <- 800
4 sigma <- 40
5 # Set seed for reproducibility
6 set.seed(123)
7 # Generate 5 plausible individual draws
8 draws <- rnorm(5, mean = mu, sd = sigma)
9 # Display the draws
10 print(round(draws, 2))
11 # Create a simple plot
12 plot(1:5, draws,
13      type = "b",
14      pch = 19,
15      xlab = "HOP Frame",
16      ylab = "Battery Life (Cycles)",
17      main = "5 Hypothetical Outcomes of Battery Charge Life")
18
19 main = "5 Hypothetical Outcomes of Battery Charge Life")
20 # Add true mean line
21 abline(h = mu, lty = 2)
22 # Add ±1 standard deviation lines
23 abline(h = mu + sigma, lty = 3)
24 abline(h = mu - sigma, lty = 3)
25 # Labels
26 text(1:5, draws, labels = round(draws, 1),
27      pos = 3, cex = 0.8)
28 legend("topright",
29       legend = c("True Mean = 800",
30                  "±1 SD = 40"),
31       lty = c(2, 3),
32       bty = "n")
```

Output:



Problem 22

Data: A battery's charge cycle life has a true mean of 800 cycles and a standard deviation of 40 cycles.

Task: List 5 plausible individual draws that could each form one frame of a Hypothetical Outcome Plot, and explain what a viewer learns from watching all 5 frames in sequence versus a single error bar.

Given

- True mean (μ) = 800 cycles
- Standard deviation (σ) = 40 cycles

A plausible individual draw should be reasonably close to 800, while allowing natural variation.

1. Five plausible individual draws

One possible set of five draws is:

760, 785, 810, 825, 840

These are plausible because they are within a reasonable range around the true mean of 800 cycles.

For example:

Frame	Individual draw
1	760 cycles
2	785 cycles
3	810 cycles
4	825 cycles
5	840 cycles

What the viewer learns from the 5 frames

Watching all 5 frames in sequence helps the viewer understand that:

- Individual battery lifetimes vary from one outcome to another.
- The outcomes tend to be centered around the true mean of 800 cycles.
- Some values are below 800 and some are above 800.
- The spread of the outcomes gives an intuitive understanding of the 40-cycle standard deviation.
- It shows that 800 cycles is the center of the distribution, not the exact value every battery will have.

3. Compared with a single error bar

A single error bar might show:

$$800 \pm 40$$

or a range from approximately:

760 to 840 cycles

This summarizes the data in one static graphic, but it does not show individual possible outcomes.

```
# Problem 22: Hypothetical Outcome Plot
# Given population parameters
mean_cycles <- 800
sd_cycles <- 40
# Generate 5 plausible individual draws
set.seed(123)
draws <- rnorm(5, mean = mean_cycles, sd = sd_cycles)
# Display the draws
cat("Five plausible individual draws:\n")
print(round(draws, 2))
# Plot the five outcomes
plot(
  1:5,
  draws,
  type = "o",
  pch = 19,
  main = "Hypothetical Outcome Plot",
```

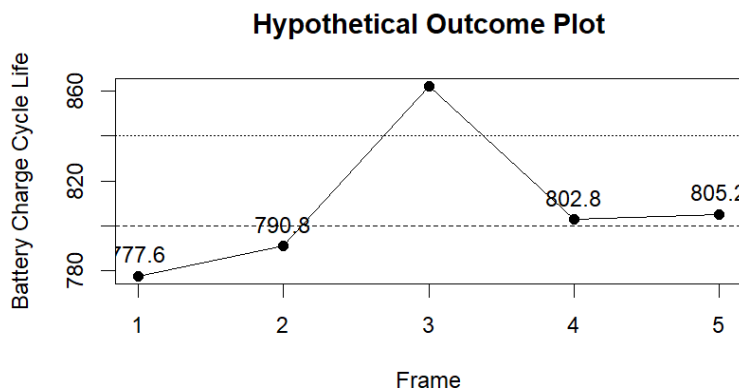
```
  main = "Hypothetical Outcome Plot",
  xlab = "Frame",
  ylab = "Battery Charge Cycle Life"
)

# Add true mean line
abline(h = mean_cycles, lty = 2)

# Add +/- 1 standard deviation lines
abline(h = mean_cycles + sd_cycles, lty = 3)
abline(h = mean_cycles - sd_cycles, lty = 3)

# Labels
text(1:5, draws, labels = round(draws, 1), pos = 3)
```

Output



Problem 23

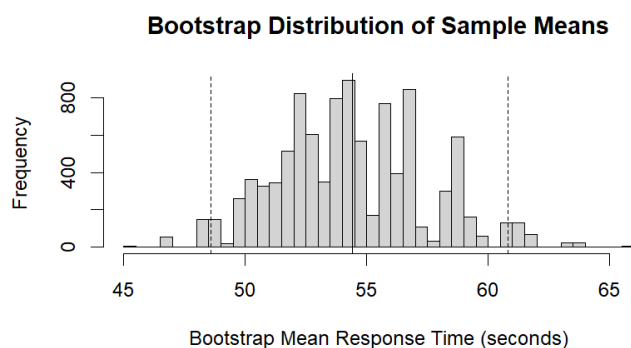
Data: A small sample of 5 response times (seconds): 45, 53, 53, 55, 66.

Task: Perform one bootstrap resample (draw 5 values with replacement) and calculate its mean, then explain how repeating this process thousands of times and taking the middle 95% of resulting means gives a confidence interval.

```
1 # Original response times
2 data <- c(45, 53, 53, 55, 66)
3 # Set seed so the result is reproducible
4 set.seed(123)
5 # Perform ONE bootstrap resample
6 bootstrap_sample <- sample(data, size = 5, replace = TRUE)
7 # Calculate bootstrap mean
8 bootstrap_mean <- mean(bootstrap_sample)
9 cat("Original data:", data, "\n")
10 cat("Bootstrap sample:", bootstrap_sample, "\n")
11 cat("Bootstrap mean:", bootstrap_mean, "seconds\n")
12 # -----
13 # Repeat bootstrap thousands of times
14 # -----
15 B <- 10000
16 bootstrap_means <- numeric(B)
17 for (i in 1:B) {
```

```
  main = "Bootstrap Distribution of Sample Means",
  xlab = "Bootstrap Mean Response Time (seconds)"
)
abline(v = CI[1], lty = 2)
abline(v = CI[2], lty = 2)
abline(
  v = mean(data),
  lty = 1
)
for (i in 1:B) {
  resample <- sample(data, size = 5, replace = TRUE)
  bootstrap_means[i] <- mean(resample)
}
# Calculate middle 95% confidence interval
CI <- quantile(bootstrap_means, c(0.025, 0.975))
cat("\nBootstrap 95% Confidence Interval:\n")
cat("Lower limit:", CI[1], "seconds\n")
cat("Upper limit:", CI[2], "seconds\n")
# -----
# Plot bootstrap distribution
# -----
hist(
  bootstrap_means,
  breaks = 30,
  main = "Bootstrap Distribution of Sample Means",
  xlab = "Bootstrap Mean Response Time (seconds)"
```

Output:



The bootstrap method repeatedly creates new samples with replacement from the

original data.

If we perform this process 10,000 times, we get 10,000 bootstrap means. These means form a bootstrap distribution.

The middle 95% of these means is obtained by taking the:

2.5th percentile

and

97.5th percentile

So,

$$95\% CI = [2.5^{th} \text{ percentile}, 97.5^{th} \text{ percentile}]$$

This interval estimates the range in which the true population mean is likely to lie.

Problem 24

Data: Website daily visitors for 4 entities: 200, 4000, 90000, 1500000 visitors.

Task: Plot this data on both a linear and a logarithmic axis, calculate the ratio between each consecutive pair of values, and explain which axis reveals the data's structure more clearly, and why.

Dataset

Entity	Daily Visitors
Entity 1	200
Entity 2	4,000
Entity 3	90,000
Entity 4	1,500,000

1. Ratio between consecutive values

Formula:

$$\text{Ratio} = \frac{\text{Current value}}{\text{Previous value}}$$

- Entity 2 / Entity 1 = $4000/200 = 20$
- Entity 3 / Entity 2 = $90000/4000 = 22.5$
- Entity 4 / Entity 3 = $1500000/90000 \approx 16.67$

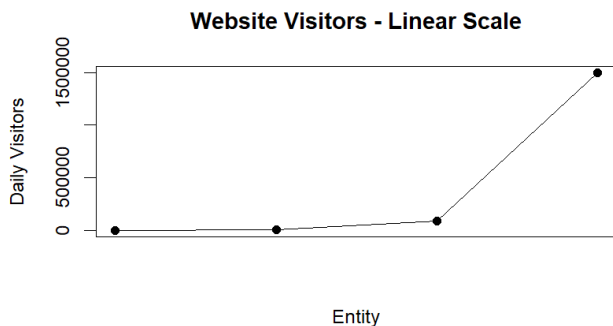
So the ratios are:

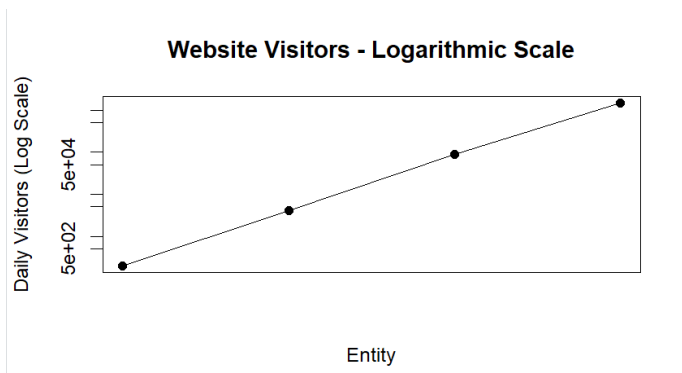
20 : 22.5 : 16.67

This shows that each entity has roughly one to two orders of magnitude more visitors than the previous one.

Python code

```
1 # Problem 24: Website Daily Visitors
2 # Data
3 visitors <- c(200, 4000, 90000, 1500000)
4 entities <- c("Entity 1", "Entity 2", "Entity 3", "Entity 4")
5 # -----
6 # 1. Plot on Linear Axis
7 # -----
8 plot(
9   1:4, visitors,
10  type = "o",
11  pch = 19,
12  xaxt = "n",
13  xlab = "Entity",
14  ylab = "Daily Visitors",
15  main = "Website Visitors - Linear Scale"
16 )
17
18 axis(1, at = 1:4, labels = entities)
19
20 # -----
21 # 2. Plot on Logarithmic Axis
22 # -----
23
24 plot(
25   1:4, visitors,
26   type = "o",
27   pch = 19,
28   log = "y",
29   yaxt = "n",
30   xlab = "Entity",
31   ylab = "Daily Visitors (Log Scale)",
32   main = "Website Visitors - Logarithmic Scale"
33 )
34
35 # -----
36 # 3. Calculate Consecutive Ratios
37 # -----
38 ratios <- visitors[-1] / visitors[-length(visitors)]
39 cat("Consecutive Ratios:\n")
40 cat("Entity 2 / Entity 1 =", ratios[1], "\n")
41 cat("Entity 3 / Entity 2 =", ratios[2], "\n")
42 cat("Entity 4 / Entity 3 =", ratios[3], "\n")
43 # -----
44 # 4. Interpretation
45 # -----
46 cat("\nInterpretation:\n")
47 cat("The ratios are approximately 20, 22.5, and 16.67.\n")
48 cat("Therefore, the number of visitors increases by roughly 17 to 23 times\n")
49 cat("between consecutive entities.\n")
50 cat("The logarithmic axis reveals the structure more clearly because the data\
```





Conclusion

The **logarithmic axis reveals the data more clearly** because the visitor counts increase by large **multiplicative factors**:

$$200 \rightarrow 4000 \rightarrow 90000 \rightarrow 1500000$$

The consecutive growth is approximately **20×**, **22.5×**, and **16.67×**. On a linear scale, the smaller values are visually compressed, while the logarithmic scale makes the relative growth easier to see.

Problem 25

Data: A report states: “There is a **45%** chance of a smartphone battery lasting over **2 years**.”

Task: Reframe this probability as a natural frequency statement, and describe how you would build a 10×10 icon array to represent it.

Python code:

```
1 # Problem 25: Simplifying a Chart for a General Audience
2 # Create sample data with 10 data points and 6 variables
3 data <- data.frame(
4   ID = 101:110,           # Irrelevant internal ID
5   Category = c("A","B","C","D","E","F","G","H","I","J"),
6   Sales = c(20, 35, 30, 45, 50, 40, 60, 55, 70, 65),
7   Profit = c(5, 8, 7, 10, 12, 9, 15, 13, 18, 16),
8   Customers = c(100, 150, 130, 180, 200, 170, 250, 220, 280, 260),
9   Cost = c(15, 27, 23, 35, 38, 31, 45, 42, 52, 49)
10 )
11 # Display the data
12 print(data)
13 # -----
14 # Original chart: includes all 6 variables
15 # -----
16 pairs(
```

```

15 # -----
16 pairs(
17   data[, c("ID", "Sales", "Profit", "Customers", "Cost")],
18   main = "Original Chart - Too Many Variables"
19 )
20 # -----
21 # Simplified chart
22 # Remove irrelevant ID and use only
23 # Category and Sales
24 # -----
25 barplot(
26   data$Sales,
27   names.arg = data$Category,
28   main = "Sales by Category",
29   xlab = "Category",
30   ylab = "Sales",
31   las = 2
32 )

```

Output:

